CLASS 8 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 3

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Which is the main cause of the rise in the CO2 level in the atmosphere? 1

(a) pollination
(b) photosynthesis
(c) rainfall
(d) burning of fossil fuels and deforestation

2. Mars is cold because it: 1

(a) is farther from the Sun and has a thin atmosphere
(b) is nearer to the Sun
(c) has a thick atmosphere
(d) has oceans

3. An object is placed very far from a convex lens. The image forms: 1

(a) at infinity **(b)** at the focus
(c) far on the same side **(d)** at the lens

4. The new Moon rises at about: 1

(a) sunset **(b)** noon **(c)** sunrise **(d)** midnight

5. The disappearance of vultures leads to a rise in: 1

(a) cows **(b)** stray dogs and diseases
(c) plants **(d)** tigers

6. Scientists get information about the interior of the Earth from: 1

(a) telescopes **(b)** seismic waves of earthquakes
(c) digging to the core **(d)** photographs of the core

7. A girl stands 1.5 m from a plane mirror. If she walks 0.5 m towards it, the distance between her and the image is: 1

(a) 2 m **(b)** 1 m **(c)** 0.5 m **(d)** 3 m

8. If the new moon is on 1 January, the next full moon will be on about: 1

(a) 15 January **(b)** 30 January **(c)** 8 January **(d)** 1 February

9. In the food web, an organism that eats both plants and animals occupies: 1

(a) no level **(b)** only the first
(c) more than one trophic level **(d)** only the third

10. At higher altitudes, breathing is difficult because: 1

(a) air is thicker
(b) air is warm
(c) air is thinner with lower oxygen amount per breath
(d) oxygen is poisonous

11. Rays parallel to the principal axis, after reflection from a concave mirror, pass through the: 1

(a) centre of the object **(b)** principal focus
(c) pole **(d)** none

12. The Hindu lunisolar calendar adds an extra month called: 1

(a) Shravana **(b)** Pausha **(c)** Magha **(d)** Adhik maas

13. Plants lose water vapour through tiny pores called: 1

(a) petals **(b)** stomata **(c)** roots **(d)** nodules

14. Which of these would you NOT regard as a renewable source of water? 1

(a) lakes **(b)** rivers
(c) fossil groundwater in deep layers **(d)** rainfall

15. Which lens can be used as a magnifying glass? 1

(a) plane glass **(b)** thick prism
(c) concave lens **(d)** convex lens of small focal length

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Why are the plastics harmful to land and water animals? 2

17. Why do mountaineers carry oxygen cylinders? 2

18. Why is the mirror used by dentists concave? 2

19. Why is the Pole Star not visible from the southern hemisphere? 2

20. If a grass field has 20000 J of energy, find the energy available to the tertiary consumer by the 10 per cent rule. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain what happens to a food web if the population of grasshoppers decreases sharply. Use frogs and snakes. 3

22. Explain why the layers of the Earth are arranged by density with the heaviest at the centre. 3

23. Two plane mirrors are at 90 degrees. A ray strikes one mirror at 30 degrees to the normal. Trace the path and the final angle between the incident and the emergent ray. 3

24. Describe how the position of the Pole Star helps a traveller find latitude. 3

OR

Explain how a wetland acts as a natural water filter. 3

25. State the effects of melting of glaciers on rivers, sea level and coastal regions. 3

OR

Why does the image formed on the retina appear inverted yet we see objects erect? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Two plane mirrors are inclined at 60 degrees with a candle placed between them. Find the number of images, and describe how they are formed. Compare with 90 degrees. **5**

OR

Describe the lunar phases through a month with the position of the Moon relative to the Sun and Earth, and the times of rising and setting. **5**

27. Explain the energy flow using a pyramid with 20000 J in producers. Compute each level, state the assumption and explain why chains are short. **5**

OR

Explain the structure of the atmosphere with heights, features and importance of each layer. **5**

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND